PLAN.md — LLM Behavioral Metrics for Interactive Fiction Logs

Overview

This project analyzes the behavior of a Large Language Model (LLM) interacting with a constrained system (Infocom-style parser).
The goal is to **quantify limitations, repetitions, and cognitive patterns** in the model's action proposals.

We treat logs as empirical traces of LLM behavior and extract metrics that reveal:

- lack of state tracking
- repetitive loops
- mismatch between reasoning and action
- exploration vs stagnation

Input Data

Log format

Logs contain sequences of:

- game outputs
- LLM thoughts (`AI thinks : ...`)
- LLM suggested commands (`AI suggests: 'COMMAND'`)

Example:

AI thinks : 'The ladder is likely the escape route' AI suggests: 'CLIMB LADDER'

Goals

1. Parse logs into structured data
2. Extract action sequences
3. Compute behavioral metrics
4. Output quantitative summaries
5. (Optional) visualize patterns

Data Model

Each step should be represented as:

Step = { "thought": str, "action": str, "result": str, }

Core Metrics

Repetition Rate

repetition_rate = repeated_actions / total_actions

Loop Detection

Detect cycles of repeated failed actions.

Action Diversity

diversity = unique_actions / total_actions

Failure Persistence

failure_retry_rate = retries_after_failure / total_failures

Thought–Action Divergence

Compare semantic similarity between thoughts and actions using embeddings.

Semantic Drift Over Time

Measure distance between consecutive thoughts.

Command Validity Approximation

Estimate syntactic validity of commands.

Exploration vs Exploitation

exploration_ratio = new_actions / total_actions

Optional Advanced Metrics

Obsession Score

obsession_score = max_consecutive_repeats / total_occurrences

Cognitive Inertia

inertia = average_repeats_after_failure

Dead-End Detection

Detect stagnation zones.

Implementation Plan

Step 1 — Parser

Extract thoughts, actions, results.

Step 2 — Structuring

Build Step objects.

Step 3 — Metrics Engine

Compute metrics.

Step 4 — Output

Export JSON summary.

Step 5 — Visualization (optional)

Graphs and timelines.

Technical Stack

- Python 3.10+
- numpy / pandas
- sentence-transformers or Ollama embeddings
- matplotlib (optional)

Expected Outcome

LLMs show:

- no persistent state
- local coherence
- poor planning
- repetitive patterns

Research Perspective

This tool exposes limitations rather than fixing them.

> LLMs generate meaning but struggle with constrained symbolic systems.

Next Steps

- analyze multiple logs
- compare models
- integrate into thesis